

Dhwani Kherawat

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CS student who builds the part of a system that only fails under a condition nobody arranges on purpose — a crash mid-write, two clients editing the same record, a drawing symbol that arrives as the wrong character. Three shipped projects, each with a test suite built to prove the failure case, not just the happy path.

PROJECTS

Kin — Shared Medication Schedule for Family Caregivers

2026

React · Node.js · MongoDB · WebSockets · Vitest · github.com/dhwanikher/kin

Real-time care-coordination app so two people caring for the same relative never give the same dose twice. Every client updates over a WebSocket the moment a dose is logged; when two requests still land together, a single atomic `findOneAndUpdate` lets exactly one win.

Verified, not assumed: swapping the atomic update for a naive read-then-write fails both concurrency tests (two-person race, five-simultaneous-taps); restoring it passes them.

- Built accessibility as a first-class requirement, not a toggle — large-text and high-contrast modes for elderly users, catching and fixing a layout bug where the largest text size pushed action buttons off-screen.

balloon — Engineering Drawing to Inspection Sheet

2026

Go · pdf.js · AS9102 / GD&T · github.com/dhwanikher/balloon

Parses toleranced dimensions out of engineering-drawing PDFs and generates a numbered First Article Inspection sheet — normally done by hand-drawing bubbles on a printout and retyping every value into a spreadsheet.

- Handles real-world PDF text corruption: the same diameter symbol arrives as $\emptyset$, $\emptyset$, or a bare ASCII letter when a symbol font's glyphs don't survive export; adjacent text runs sharing a baseline are stitched back into one callout before parsing.
- Single Go binary with pdf.js vendored — no build step, no network call ever leaves the machine with the drawing.

stow — Embedded Key-Value Store Tested by Simulated Power Loss

2026

Go · write-ahead log · memtable · github.com/dhwanikher/stow

A durability contract can't be verified by testing on a real disk, since the failure that matters — power loss mid-write — can't be arranged on demand. Built a filesystem interface with a mock implementation that triggers a simulated crash at every durability-relevant operation a workload performs.

43 crash points per workload, each replayed with a clean crash and three torn-write variants — about 170 recovery scenarios verified per test run.

EDUCATION

Scaler School of Technology — B.Tech, Computer Science (degree awarded by BITS Pilani), Bengaluru

2026 – 2029

TECHNICAL SKILLS

Languages: JavaScript, Go, HTML/CSS, SQL

Frontend: React, Vite, Service Workers, ARIA / accessibility

Backend: Node.js, MongoDB, WebSockets, REST APIs

Systems & practice: Write-ahead logs, concurrency, Git, Vitest, Go testing